

Suggestions for physics.

#### 1. For Rlc, Rc circuit

- calculate total impedance,
- total impedances for resistor, conductor, capacitor
- what is meant by resonance by Rlc circuit
- power and power factor.
- 1-2 problems will be given where u have to calculate the value of impedences numerically ( phasor algebra)
- current divider rule and voltage divider rule in Ac circuit ( using phasor algebra)
- thevenins theorem--> calculate  $Z_{th}$

#### 2. Magnetism

- all the definitions( Magnetic flux, flux density..etc) along with unit.
- magnetic effect of electric current
- value of force on a charge inside a magnetic field -->  $F = qVB\sin(\theta)$
- value of force on a current carrying wire
- what is drift velocity
- what is meant by torque, and how much force is necessary to develop a torque,  $T = MB\sin(\theta)$
- Faraday's law and lenz law of induction
- self induction and mutual induction
- how to find the self inductance of a solenoid
- Biot Savarts law and Amperes law ( derivation and application)
- two applications of Biot Savarts law- long current carrying wire and two parallel conductors
- two applications of Amperes law- long straight wire and magnetic field acting for a solenoid
- **PRACTICE PROBLEMS FROM BIOT SAVARTS LAW AND AMPERES LAW---> FORMULA**

#### 3. Types of magnetism

- ferromagnetic, diamagnetic
- curie temperature
- hysteresis ( draw the hysteresis curve)
- soft magnetic materials and hard magnetic materials
- saturated magnetism
- retentivity
- hysteresis loss and Eddy current loss\*\*\*
- coresivity and residue magnetism

#### 4. Magnetic circuit

- flow of magnetic flux
- coil operates a magnetic circuit